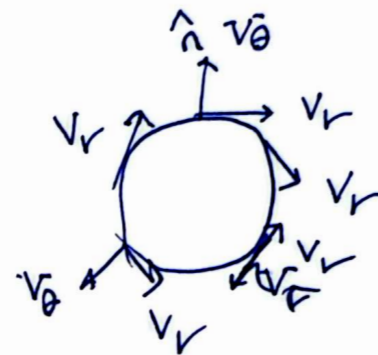
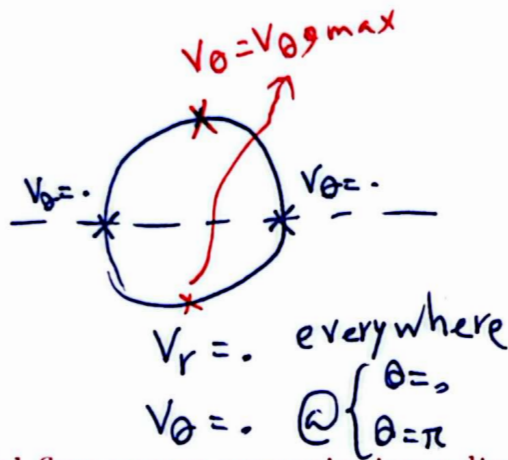




Potential flow over a non-spinning cylinder

stagnation means

$$V_r = V_\theta = 0$$

$$V_r = 0 \rightarrow r = R$$

$$V_\theta = 0 \text{ if } \theta = 0, \pi$$

$$(\sin \theta) = 0$$

C_p over
the cylinder

$$\psi = U_\infty r \sin \theta \left(1 - \frac{B}{2\pi U_\infty r^2} \right) \quad (2.106)$$

$$\begin{cases} v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U_\infty \cos \theta \left[1 - \frac{B}{2\pi U_\infty r^2} \right] = 0 \end{cases} \quad (2.107)$$

$$v_\theta = -\frac{\partial \psi}{\partial r} = -U_\infty \sin \theta \left[1 + \frac{B}{2\pi U_\infty r^2} \right] \quad (2.108)$$

$$\text{Stagnation pt. } |\vec{V}| = 0 \rightarrow r = R = \sqrt{\frac{B}{2\pi U_\infty}} \quad (2.109)$$

$$C_p = 1 - \left(\frac{U}{U_\infty} \right)^2 = 1 - \left(\frac{\sqrt{v_r^2 + v_\theta^2}}{U_\infty} \right)^2 \quad (2.110)$$

$$C_p = 1 - 4 \sin^2 \theta \quad (2.111)$$

$$\vec{V} \cdot \vec{n} = 0$$

$$\vec{V} = V_r \hat{e}_r + V_\theta \hat{e}_\theta$$

$$\hat{n} = \hat{e}_\theta$$

$$V_r = 0 \Rightarrow S_0$$

$$\vec{V} \cdot \vec{n} = V_r \cdot \hat{e}_\theta = 0$$

$$C_p = 1 \text{ @ } \theta = 0, \pi$$

$$C_p = -3 \text{ @ } \theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$C_p = 0 \text{ @ } \sin \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$V_r = 0 \Rightarrow 1 - \frac{B}{2\pi U_\infty r^2} = 0$$

$$r^2 = \frac{B}{2\pi U_\infty} \text{ or } r = \sqrt{\frac{B}{2\pi U_\infty}} \begin{cases} \theta = \frac{\pi}{6} \\ \theta = \frac{5\pi}{6} \end{cases}$$



$$C_p = 0 \Rightarrow \frac{1}{4} = \sin^2 \theta \rightarrow$$

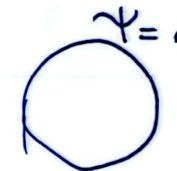
R

$$R = \sqrt{\frac{B}{2\pi U_\infty}}$$

Potential flow over a non-spinning cylinder

$$\psi = U_\infty r \sin \theta \left[1 - \left(\frac{R}{r} \right)^2 \right] \quad (2.112)$$

$$\psi(r=R) = \underline{0} \rightarrow \text{surface streamline} \quad (2.113)$$



Note, that it just happens that $\psi = 0$ is the surface streamline. There is no other significance to this.

- Stagnation point $\rightarrow \theta = 0 \text{ \& } \pi, r = R$. (Show as an ungraded homework)
- Max velocity $\rightarrow r = R, \theta = \pi/2 \text{ \& } 3\pi/2, v_r = 0, v_\theta = 2U_\infty$ (Show as an ungraded homework)
- At $r = R, v_r = 0$ and $v_\theta = -2U_\infty \sin \theta$ (Show as an ungraded homework)

$$v_\theta = -2U_\infty \sin \theta$$

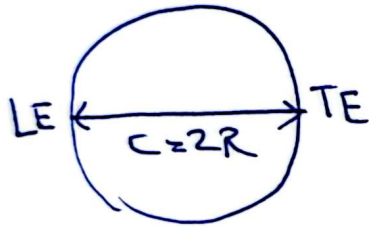
over
the
cyl ($r=R=\sqrt{\frac{B}{2\pi U_\infty}}$)

$$v_r = 0$$

$$C_d = \frac{1}{c} \int_{LE}^{TE} (C_{pu} - C_{pl}) dy = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta$$

chord : $c = 2R$

$$C_l = \frac{-1}{2} \int_0^{2\pi} C_p \sin \theta d\theta$$



Potential flow over a non-spinning cylinder

- $C_l = \frac{1}{c} \int_{le}^{te} (C_{pl} - C_{pu}) dy = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta = 0$. Can also be determined by observation of C_p distribution plot. There is top/bottom symmetry hence there is no net force in the y direction and thus no lift.
- $C_d = \frac{1}{c} \int_{le}^{te} (C_{pu} - C_{pl}) dx = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta = 0$. Can also be determined by observation of C_p distribution plot. There is left/right symmetry hence there is not net force in the x direction and thus no drag.

$$C_l = \frac{1}{c} \int_{LE}^{TE} (C_{pl} - C_{pu}) dx$$

$$C_d = \frac{1}{c} \int_{le}^{te} (C_{pu} - C_{pl}) dy \quad (2.114)$$

$$y = R \sin \theta \rightarrow dy = R \cos \theta d\theta, \quad c = 2R \quad (2.115)$$

$$C_d = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta, \quad \text{Similarly } C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta \quad (2.116)$$

Potential flow over a non-spinning cylinder

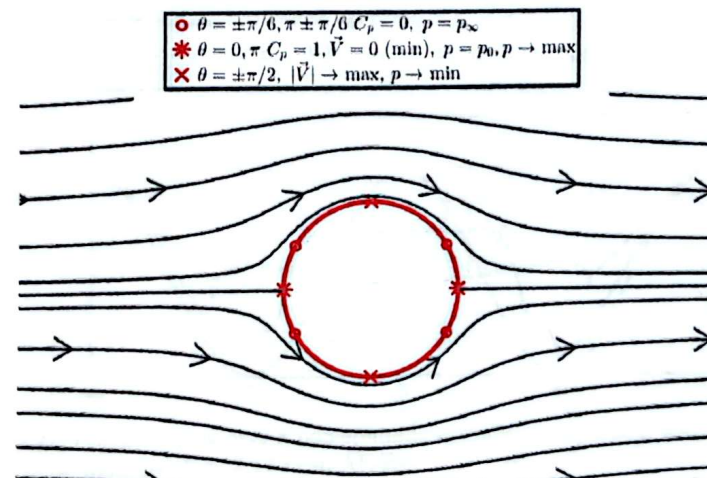


Figure 2.21: Potential flow over a non-spinning cylinder. Based on image from FA-JA

Subcritical
flow



$C_p \approx \text{cte}$ in
separation region

Potential flow over a non-spinning cylinder

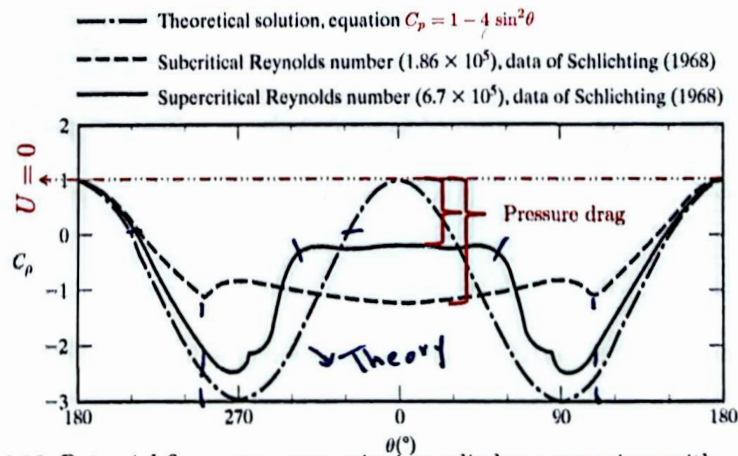
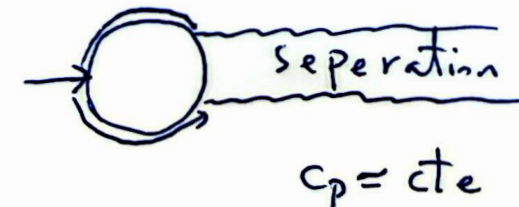


Figure 2.22: Potential flow over a non-spinning cylinder, comparison with experiments, modified Fig. 3.17 from AE-BC

"super-critical"



Potential flow over a spinning cylinder (TSec. 3.15)

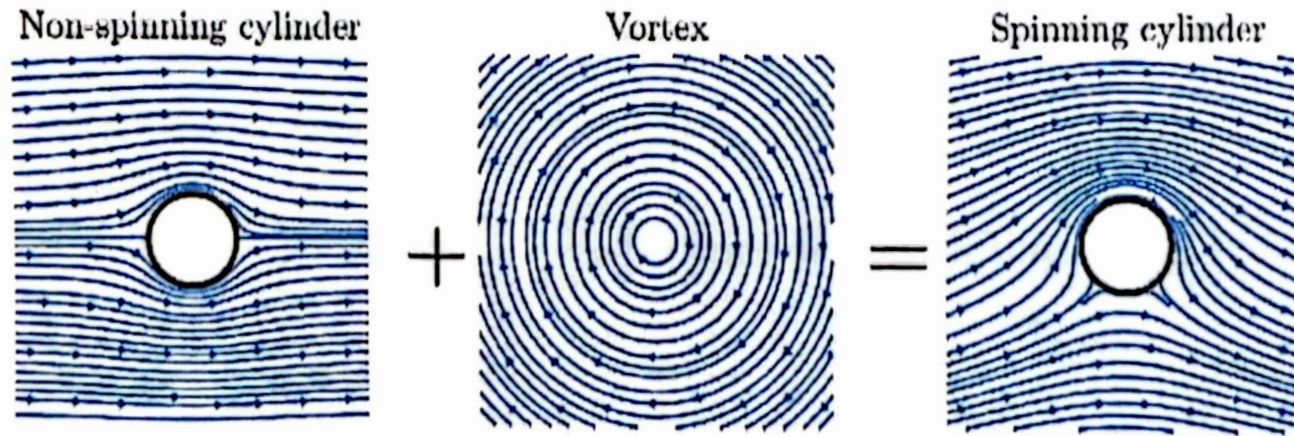


Figure 2.23: Potential flow over a spinning cylinder. Based on image from FA-JA

$$\psi(r, \theta) = \psi_{\text{free-stream}} + \psi_{\text{doublet}} + \psi_{\text{vortex}} \quad (2.117)$$

$$= \psi_{\text{non-spinning cylinder}} + \psi_{\text{vortex}} \quad (2.118)$$

Potential flow over a spinning cylinder

$$\psi(r, \theta) = \underbrace{U_{\infty} r \sin \theta \left[1 - \left(\frac{R}{r} \right)^2 \right]}_{\psi_{\text{non-spin}}} + \underbrace{\frac{\Gamma}{2\pi} \ln \frac{r}{R}}_{\psi_{\text{vortex}}} \quad (2.119)$$

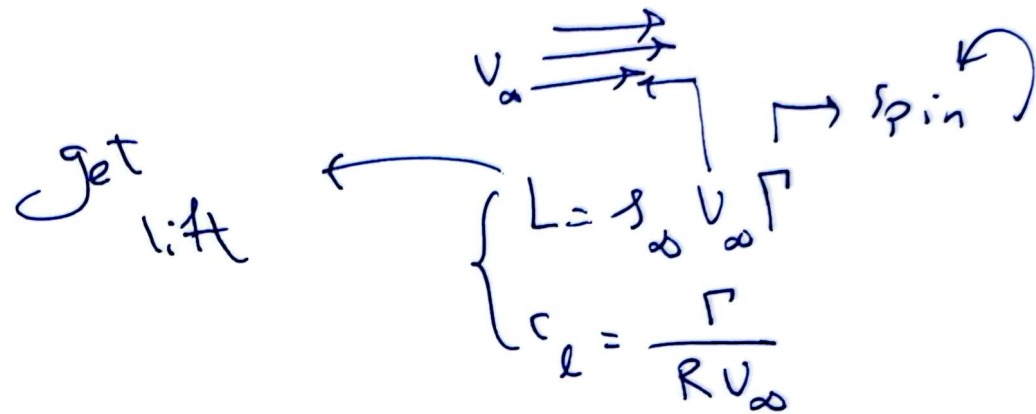
$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U_{\infty} \cos \theta \left[1 - \frac{R^2}{r^2} \right] \rightarrow \text{same as non-lifting} \quad (2.120)$$

$$v_{\theta} = -\frac{\partial \psi}{\partial r} = \underbrace{-U_{\infty} \sin \theta \left[1 + \frac{R^2}{r^2} \right]}_{\text{non-lifting}} \underbrace{\left(-\frac{\Gamma}{2\pi r} \right)}_{\text{Extra term due to vortex}} \quad (2.121)$$

appears in v_{θ}

$$C_p = \underbrace{1 - 4 \sin^2 \theta}_{\text{non-lifting}} + \frac{-2\Gamma \sin \theta}{\pi R U_{\infty}} - \left(\frac{\Gamma}{2\pi R U_{\infty}} \right)^2 \quad (2.122)$$

$$C_p = 1 - \left(\frac{v}{U_{\infty}} \right)^2 = 1 - \left(\frac{\sqrt{v_r^2 + v_{\theta}^2}}{U_{\infty}} \right)^2$$



Potential flow over a spinning cylinder

"Magnus effect"

$$C_d = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta \, d\theta = 0 \quad (2.123)$$

$$C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta \, d\theta = \frac{\Gamma}{R U_\infty} \quad \text{after some algebra} \quad (2.124)$$

$$l = \frac{1}{2} C_l \rho_\infty U_\infty^2 c = \frac{1}{2} \frac{\Gamma}{R U_\infty} \rho_\infty U_\infty^2 2R = \rho_\infty U_\infty \Gamma \quad (2.125)$$

$$C_d = 0, \quad C_l = \frac{\Gamma}{R U_\infty}, \quad l = \rho_\infty U_\infty \Gamma$$

"Spinning" the cylinder causes an asymmetrical force due to Γ which in turn causes lift - Magnus effect

Potential flow over a spinning cylinder

- ▶ Stagnation point $\rightarrow \theta = \sin^{-1} -\Gamma/4\pi RU_\infty$, $r = R$. (Show as an ungraded homework)
- ▶ Max velocity $\rightarrow r = R$, $\theta = \pi/2$ (Show as an ungraded homework)
- ▶ Stagnation streamline is still $\psi = 0$ (Show as an ungraded homework)
- ▶ At $r = R$, $v_r = 0$ and $v_\theta = -2U_\infty \sin\theta - \Gamma/2\pi R$ (Show as an ungraded homework)
- ▶ $C_l = \Gamma/RU_\infty$.
- ▶ $C_d = 0$. Can also determine by observation of C_p distribution plot. There is left/right symmetry hence there is not net force in the x direction and thus no drag.

Note: Since the spinning cylinder produces lift sometimes we refer to it as the lifting cylinder and the non-spinning cylinder as a non-lifting cylinder

Potential flow over a spinning cylinder

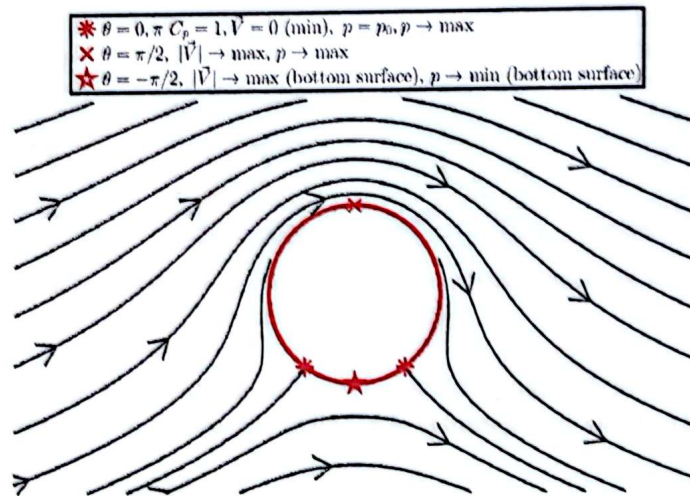
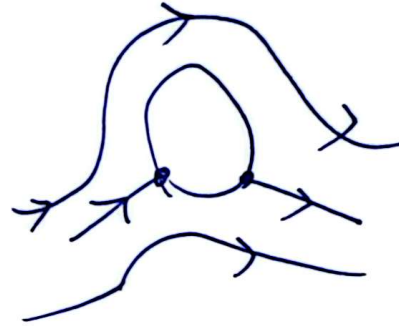


Figure 2.24: Potential flow over a spinning cylinder



Potential flow over a spinning cylinder

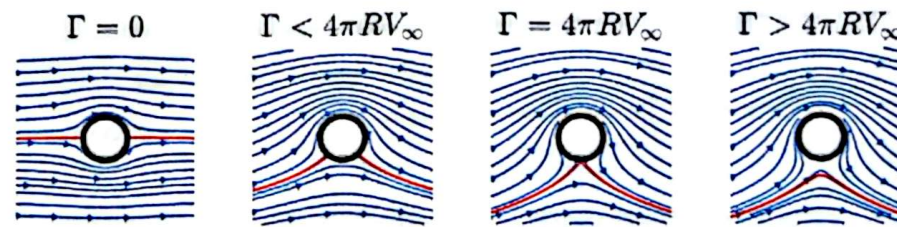


Figure 2.25: Potential flow over a spinning cylinder. Based on image from FA-JA

Γ determines the locations of the **stagnation** points. Γ is also directly proportional to how fast the cylinder is spinning and also to the lift.

Ex.1) A Uniform flow in x-direction
+ Sink ($\Lambda < 0$) at the origin

a) Find location of stagnation points

b) plot streamlines

c) write streamline Eq.ⁿ

Solution:

$$\psi = \psi_{\text{uniform}} + \psi_{\text{sink}} = U_{\infty} r \sin \theta + \frac{\Lambda}{2\pi} \theta$$

$$\phi = U_{\infty} r \cos \theta + \frac{\Lambda}{2\pi} \ln r$$

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U_{\infty} \cos \theta + \frac{\Lambda}{2\pi r}$$

$$V_{\theta} = -\frac{\partial \psi}{\partial r} = -U_{\infty} \sin \theta$$

a) stagnation points ($V_r = V_{\theta} = 0$), $V_{\theta} = 0 \rightarrow \theta = 0, \pi$

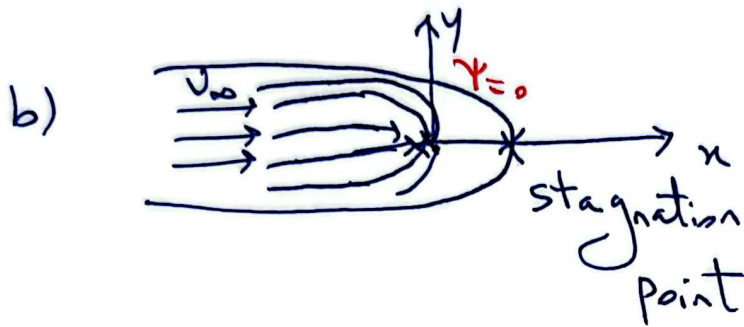
$$V_r = 0 \Rightarrow \begin{cases} U_{\infty} \cos \theta = -\frac{\Lambda}{2\pi r} \\ \theta = 0, \pi \\ \Rightarrow U_{\infty} = \frac{-\Lambda}{2\pi r} \end{cases}$$

$$\Rightarrow r = \frac{-\Lambda}{2\pi U_{\infty}}, \Lambda < 0$$

$$\rightarrow r = \frac{|\Lambda|}{2\pi U_{\infty}}$$

①

plot streamlines



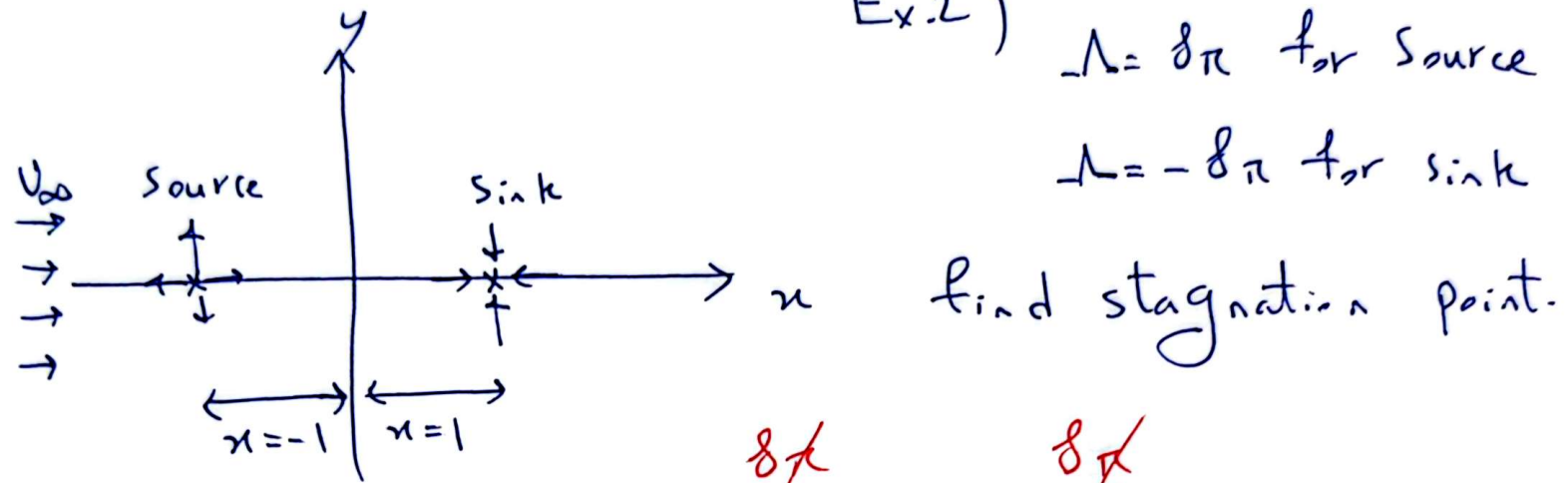
$$\left\{ r = \frac{|\Lambda|}{2\pi U_\infty}, \theta = 0 \right.$$

c) $\Psi = U_\infty r \sin \theta + \frac{\Lambda}{2\pi} \theta$

at stagnation point, $\theta = 0$

$$\Psi_{\text{stag.}} = 0 + 0 = 0 \Rightarrow \Psi_{\text{stag.}} = 0$$

Ex.2) $\Lambda = 8\pi$ for Source
 $\Lambda = -8\pi$ for Sink

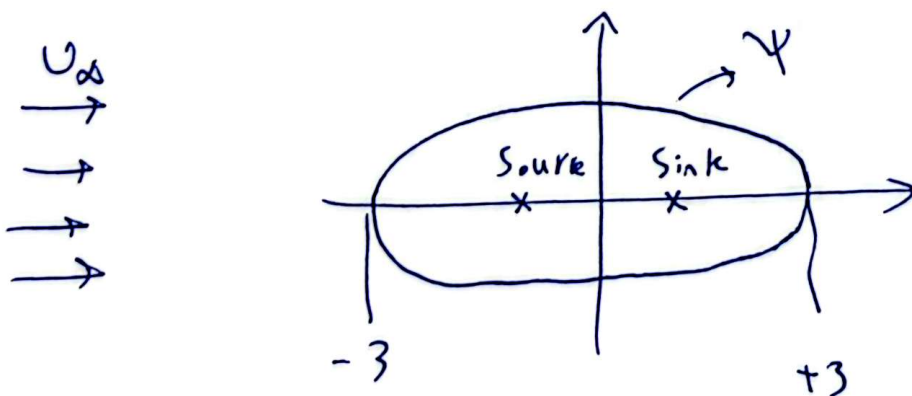


Solution: $u(x) = U_{\infty} + \frac{\cancel{8\pi} \Lambda}{2\cancel{\pi}(x+1)} + \frac{\cancel{8\pi} \Lambda}{2\cancel{\pi}(x-1)} = U_{\infty} + \frac{4}{x+1} - \frac{4}{x-1}$

$\left\{ \begin{array}{l} \overline{x-x_0} \\ x_0 = -1 \end{array} \right\}$

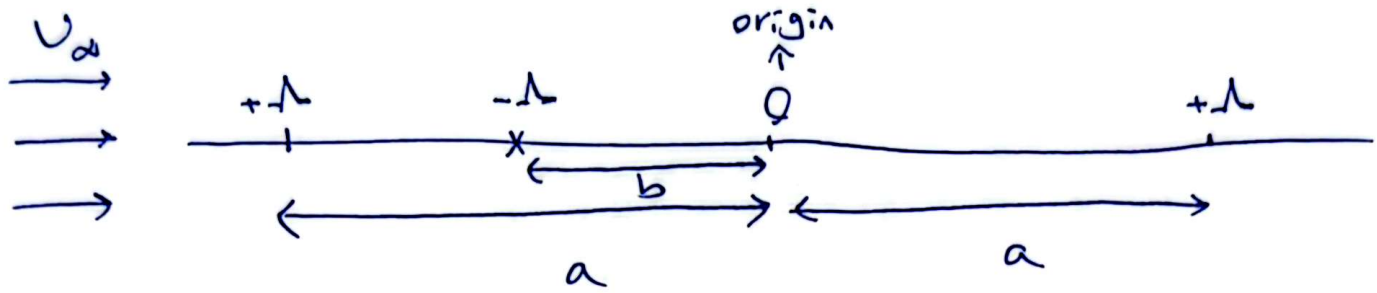
$$U_{\infty} + \frac{4(x-1) - 4(x+1)}{(x+1)(x-1)} = 0 \Rightarrow \begin{cases} U_{\infty} - \frac{8}{x^2-1} = 0 \\ U_{\infty} = 1 \end{cases}$$

$$\Rightarrow 1 - \frac{8}{x^2-1} = 0 \Rightarrow x^2 - 9 = 0 \Rightarrow x = \pm 3$$



(2)

Ex.3 We have this configuration:



$$a = \sqrt{2} b$$

- Find a relation between U_∞ , Λ , and b .
- Find stagnation points.
- plot streamlines.

Solution:

a) Velocity :

$$U_\infty + \frac{\Lambda}{2\pi(x+a)} - \frac{\Lambda}{2\pi(x+b)} + \frac{\Lambda}{2\pi(x-a)}$$

stagnation points

$$(x-x_s)$$

$$x_s = -a$$

$$x_s = -b$$

$$x_s = a$$

$$\text{Velocity} = 0 \Rightarrow \begin{cases} U_\infty + \frac{\Lambda}{2\pi(x+a)} - \frac{\Lambda}{2\pi(x+b)} + \frac{\Lambda}{2\pi(x-a)} = 0 \\ a = \sqrt{2} b \end{cases}$$

The question says that 0 is the first stagnation point, so $V=0$ @ 0

If $x=0$ is the first stagnation point, so
put $x=.$ in V relation,

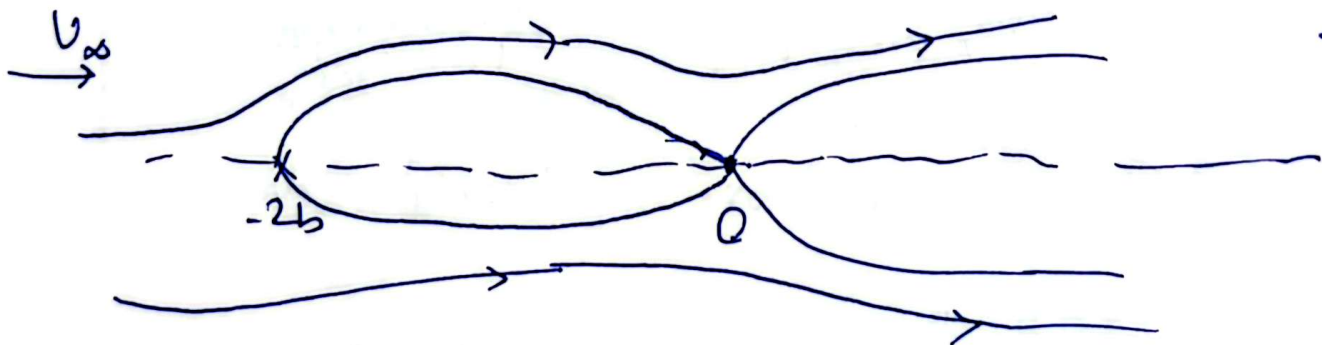
$$U_{\infty} + \frac{\cancel{\Lambda}}{2\pi a} - \frac{\Lambda}{2\pi b} - \frac{\cancel{\Lambda}}{2\pi a} = 0 \implies \text{Solve } \boxed{U_{\infty} = \frac{\Lambda}{2\pi b}}$$

put this U_{∞} in V relation:

$$\frac{\cancel{\Lambda}}{2\pi b} + \frac{\cancel{\Lambda}}{2\pi(x+a)} - \frac{\Lambda}{2\pi(x+b)} + \frac{\cancel{\Lambda}}{2\pi(x-a)} = 0$$

$$\implies \frac{1}{b} + \frac{1}{x+a} - \frac{1}{x+b} + \frac{1}{x-a} = 0 \quad \text{Simplify:}$$

$$\implies x^2(x+2b) = 0 \implies \begin{cases} x=0 \text{ (double root)} \\ x=-2b \end{cases}$$



Problem 2

Consider stagnation point potential flow over a flat plate as shown below on the left. The stream function for this flow is given by

$$\psi = Axy$$

where A is a constant. The origin O here is the stagnation point. (Note - You can use the stream function above to prove this. You may verify this if you desire additional practice). To this stagnation point flow we add a source of strength Λ which generates potential flow over bump as shown below on the right. This would simulate flow inside a heat exchanger, for example, when there is some form of mineral deposition. Another example would be a jet exhaust impinging on an object during take off.

- Determine the bump height h .
- Determine the surface streamline ψ_s .
- Write a script to plot the streamlines. Use $A = \Lambda = 1$. Plot ψ_s using a different color. Highlight the location of the stagnation point.

Hint: ψ_s has to pass through the stagnation point.

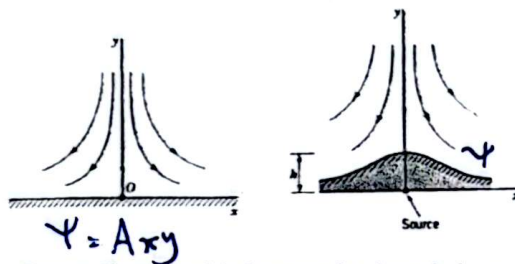


Figure 1: Illustration of the flow over a flat plate and a bump.

a) $\psi = \psi_{stg} + \psi_{source}$

$$\psi = Axy + \frac{\Lambda \theta}{2\pi}$$

If $x = r \cos \theta$

$$y = r \sin \theta$$

$$\text{Then } \psi = Ar \cos \theta \sin \theta + \frac{\Lambda \theta}{2\pi}$$

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = Ar(\cos^2 \theta - \sin^2 \theta) + \frac{\Lambda}{2\pi r}$$

$$= Ar \cos 2\theta + \frac{\Lambda}{2\pi r}$$

if $r = h$

$$V_r = 0 = Ah \cos 2\theta + \frac{\Lambda}{2\pi h}$$

$$\theta_{stag} = \frac{\pi}{2}$$

$$0 = Ah \cos 2 \cdot \frac{\pi}{2} + \frac{\Lambda}{2\pi h}$$

$$Ah = \frac{\Lambda}{2\pi h}$$

$$h^2 = \frac{\Lambda}{2\pi A}$$

$$\therefore h = \sqrt{\frac{\Lambda}{2\pi A}}$$

b) From above, $\psi_{stag} = Ar \cos \theta \sin \theta + \frac{\Lambda \theta}{2\pi}$ when $h =$ and $\theta = \frac{\pi}{2}$

$$= Ar^2 \cos \theta \sin \theta + \frac{\Lambda \theta}{2\pi}$$

$$= A \left(\sqrt{\frac{\Lambda}{2\pi A}} \right)^2 \cos \left(\frac{\pi}{2} \right) \sin \left(\frac{\pi}{2} \right) + \frac{\Lambda \left(\frac{\pi}{2} \right)}{2\pi}$$

$$\psi_{stag} = \frac{\Lambda}{h}$$

$$\therefore \psi_s = Ar^2 \cos \theta \sin \theta + \frac{\Lambda \theta}{2\pi} = \frac{\Lambda}{h}$$

$$\psi = Axy$$

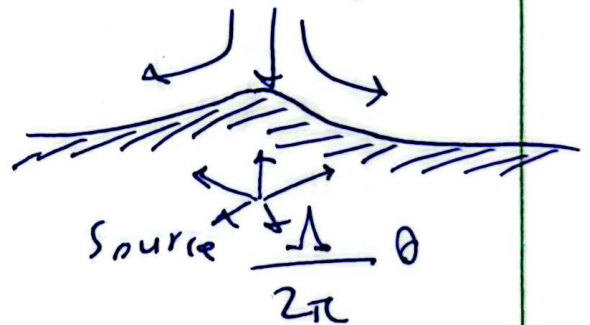
$$(.,.) \rightarrow s.p. ?$$

$$u = \frac{\partial \psi}{\partial y} = Ax$$

$$V = -\frac{\partial \psi}{\partial x} = -Ay$$

$$x=y=0 \rightarrow u=V=0$$

$$\psi = Axy$$



$$\psi = \psi_{stg} + \psi_{source}$$

$$= Axy + \frac{\Lambda}{2\pi} \theta$$

$$= Ar \cos \theta \sin \theta + \frac{\Lambda}{2\pi} \theta$$

$$= Ar^2 \cos \theta \sin \theta + \frac{\Lambda}{2\pi} \theta$$

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} =$$

$$Ar(\cos^2 \theta - \sin^2 \theta) + \frac{\Lambda}{2\pi r}$$

$$= Ar \cos 2\theta + \frac{\Lambda}{2\pi r}$$

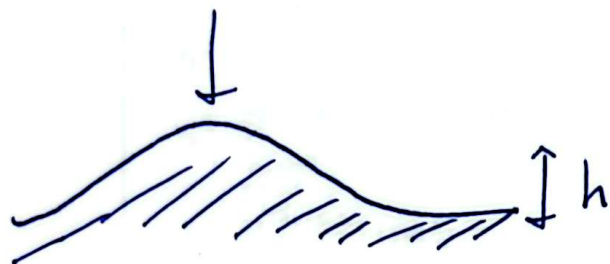
over the surface $V_r = 0$
 $r = h$ (height of hills)

$$\Rightarrow Ah \cos 2\theta + \frac{\Lambda}{2\pi h} = 0$$

Prove it by putting

$$\theta \text{ of } h = \frac{\pi}{2}$$

$$V_\theta = 0$$



$$\Rightarrow Ah(-1) = -\frac{\Lambda}{2\pi h}$$

$$\Rightarrow h^2 = \frac{\Lambda}{2\pi A}$$

$$\Rightarrow h = \sqrt{\frac{\Lambda}{2\pi A}}$$

stagnation streamfunction:

$$\psi = Ar^2 \cos \theta \sin \theta + \frac{\Lambda \theta}{2\pi}, \quad h = \sqrt{\frac{\Lambda}{2\pi A}} = r$$

$$\theta = \frac{\pi}{2}$$

put h & θ in ψ .

$$\Rightarrow \psi = A \left(\sqrt{\frac{\Lambda}{2\pi A}} \right)^2 \cos \frac{\pi}{2} \sin \frac{\pi}{2} + \frac{\Lambda}{2\pi} \frac{\pi}{2} = \frac{\Lambda}{4}$$

